

Informativeness under Model Uncertainty: Shadow Prices and Ridge Penalties

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Introduction

Motivation: Model Uncertainty in Restrictions

- **Multiple restrictions $\Rightarrow$ model uncertainty**
 - ▶ Economic theory often imposes a *system* of restrictions on parameters.
 - ▶ Researchers face many candidate restrictions without knowing which are empirically relevant.

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- **Sensitivity to restriction choice**

- ▶ Empirical conclusions depend on which restrictions are imposed.
- ▶ Natural shift: from selecting one model $\rightarrow$ assessing *relative informativeness of competing models*.
 - ★ Realm of the model averaging avoids $\{0, 1\}$ selection – but optimal weights to different models.
 - ★ Ridge: model averaging; LASSO: model selection

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- **Approximate empirical relevance of theory**

- ▶ Including noisy or weak restrictions complicates estimation:
 - ★ imposing all restrictions may introduce bias if some are misspecified,
 - ★ discarding them sacrifices efficiency when they are informative.
- ▶ In addition, noisy or weak restrictions inflate multiplicity, reducing the power of standard validity tests.

Framework

- Consider $m_1(\text{model}) + m_2(\text{nuisance}) + e$
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- Define the quadratic restriction measure

$$h(\theta) = g(\theta)' \Sigma^{-1} g(\theta).$$

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- Define the quadratic restriction measure

$$h(\theta) = g(\theta)' \Sigma^{-1} g(\theta).$$

- For each tolerance level $c \in \mathcal{C} \subset (0, c_0]$, the constrained estimator solves

$$\hat{\theta}(c) \in \arg \min_{\theta \in \Theta} \phi_n(\theta) \quad \text{s.t.} \quad h(\theta) \leq c.$$

- ▶ $\phi(\theta)$ is a loss function.
- ▶ Small c enforces the restrictions tightly, while larger c allows greater deviation.

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Research Questions and Implications

- **Research questions**

- ▶ Under model uncertainty, how can we consistently estimate the parameters as in the unconstrained model?
- ▶ How can we assess the *empirical relevance* of each restriction?

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- ▶ How can we assess the *empirical relevance* of each restriction?

● Implications

- ▶ Robust estimation in the presence of uncertain or misspecified restrictions (through the optimized tolerance for restrictions misspecifications)
- ▶ A statistical guidance on assessing empirical relevance of each restriction

Introduction

Challenges

- **Setup: finite set of restrictions**

- ▶ Theoretical restrictions are incorporated via constrained optimization
- ▶ Allow for *restriction misspecification* through a tolerance parameter

- **Challenges**

1. Regularization bias from (possibly misspecified) restrictions
2. How to assess the empirical relevance of individual restrictions

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Issue 1: Regularization Bias

• Challenges

- ▶ Constrained estimation targets a *pseudo-true parameter*
- ▶ Bias arises when restrictions are misspecified
- ▶ How does c govern the degree of misspecification?

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Issue 1: Regularization Bias

● Challenges

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- ▶ How does c govern the degree of misspecification?

● Our framework

- ▶ c -level (tolerance selection):
 - ★ Introduce a tolerance parameter c to allow controlled deviation from restrictions
 - ★ Choose c by minimizing a Stein-type risk, balancing bias and variance
 - ★ The optimized c determines how strongly restrictions are enforced by the data
- ▶ θ -level (estimation and inference):
 - ★ Further refinement via KKT conditions to construct a debiased estimator

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Issue 2: Empirical Relevance

- **Challenge**

- ▶ How can we assess and decompose the empirical relevance of individual restrictions?

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- **Challenge**

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- **Our framework**

- ▶ Introduce *individual shadow prices (ISP)*
 - ★ Defined as directional derivatives with respect to each restriction
 - ★ Measure the marginal gain in the model fit from relaxing one restriction, holding others fixed
 - ★ Proportional to the global shadow price $\Rightarrow$ provide a relative scale for ranking restrictions
- ▶ Provide statistical guidance for separating signal from noise
 - ★ Detect a structural break in ordered ISP magnitudes
 - ★ Use a bootstrap procedure to account for sampling variability

Introduction

Key Findings

• Estimation

- ▶ Consistently estimates the parameters in the unconstrained model
- ▶ Nominal coverage is recovered to the target level

• Individual shadow prices (ISP)-based inference

- ▶ ISP formula: $ISP_j = 2\lambda \text{sign}(g_j) [\Sigma^{-1}g]_j$, where λ is a global shadow price, g_j is the j th restriction, and Σ is a known credibility matrix.
- ▶ ISPs quantify the empirical relevance of each restriction in improving fit.
- ▶ Bootstrap-based plateau rule provides a data-driven separation between signal and noise.

• Implications

- ▶ Estimation robust to model uncertainty via controlled restriction misspecification, using tolerance optimization and debiasing
- ▶ Inference on the empirical relevance of individual restrictions through shadow-price interpretation

Introduction

Related literature and contribution

1. Restrictions from Economic Theory

- Theory imposes cross-parameter/equilibrium restrictions directly in the model
 - ▶ Growth models (Solow, 1956; Swan, 1956; Mankiw et al., 1992; Ertur and Koch, 2007)
 - ▶ Rational expectations (Hansen and Sargent, 1980)
 - ▶ Auction models (Guerre et al., 2000)
- Reduced-form models and test theory-implied restrictions
 - ▶ VAR present-value tests (Campbell and Shiller, 1987)
 - ▶ Revealed preference and shape-restriction tests in demand analysis (Blundell, 2005)
 - ▶ Asset pricing tests that assess zero-alpha conditions or bounds on risk aversion (Fama and MacBeth, 1973; Mehra and Prescott, 1985)

Our framework

- Data-driven ranking and selection of restrictions based on empirical relevance
- Flexible framework accommodating linear and nonlinear restrictions
- Can accommodate moment conditions

Introduction

Related literature and contribution

2. Model Uncertainty, Ambiguity, and Misspecification

- Model uncertainty in restrictions
 - ▶ Multiple candidate restrictions with unknown relevance
 - ▶ Uncertainty extends to both parameters and restrictions
- Related literature
 - ▶ Model averaging and ambiguity frameworks allow deviations from a benchmark model
 - ▶ Robustness assessed via neighborhoods of distributions or moment conditions
 - ▶ E.g., Maasoumi (1978), Gospodinov et al. (2013, 2016), Bonhomme and Weidner (2022), Christensen and Connault (2023)
- Our framework
 - ▶ Parameterizes misspecification via a tolerance level c
 - ▶ Optimizes c rather than fixing it $\Rightarrow$ balances fit and restriction adherence
 - ▶ Key distinctions
 - ★ Misspecification treated as a *decision variable*, not an ex post measure
 - ★ Explicit bias–variance tradeoff via risk-optimization of c
 - ★ Lagrange multipliers interpreted as ISPs, decomposing restriction-level contributions

Introduction

Related literature and contribution

3. Inference and regularization in Machine Learning (ML)

- Connection to regularization
 - ▶ Duality between constrained optimization and penalized estimation (Hoerl and Kennard, 1970; Tibshirani, 1996; Zou and Hastie, 2005)
 - ▶ Tolerance parameter controls deviation from restrictions (regularization strength)
- Key difference
 - ▶ ML: tuning parameters chosen for prediction
 - ▶ Our approach: tolerance selected via Stein-type risk for statistical optimality
- Inference contribution
 - ▶ Our method enables valid inference for structural parameters in the presence of regularization bias.
 - ▶ ISP provides inference on the empirical relevance of restrictions
- Relation to ML interpretation tools
 - ▶ Unlike SHAP (Lundberg and Lee, 2017), ISP evaluates *restriction relevance*
 - ▶ Naturally incorporates cross-parameter relationships from economic theory

Introduction

Outline

1. Roadmap of theory
2. Estimation
3. Inference
4. Monte Carlo Simulations
5. Empirical Application

Roadmap

- We treat theory-based restrictions as *soft constraints*, governed by an optimized tolerance parameter.
 - ▶ Instead of posing $g(\theta) = 0$, allow deviations through the constraint $h(\theta) \leq c$, letting the data determine the extent of enforcement.
 - ▶ The tolerance parameter is selected via a *Stein-type analytical risk criterion*, balancing variance reduction and misspecification-induced bias.

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 - ▶ The tolerance parameter is selected via a *Stein-type analytical risk criterion*, balancing variance reduction and misspecification-induced bias.
- The inner problem yields a constrained estimator and an associated *shadow price*, with a clear economic interpretation.
 - ▶ A shadow price interpretation: marginal gain in fit from relaxing the restriction.
 - ▶ Importantly, we decompose the global shadow price into *individual shadow prices (ISP)*, providing restriction-specific measures of empirical relevance.
 - ▶ We propose a *bootstrap-based plateau rule* distinguishes signal from noise across multiple restrictions.

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 - ▶ Importantly, we decompose the global shadow price into *individual shadow prices (ISP)*, providing restriction-specific measures of empirical relevance.
 - ▶ We propose a *bootstrap-based plateau rule* distinguishes signal from noise across multiple restrictions.
- Finally, we debias the constrained estimator to restore correct centering for inference on the parameters in the unconstrained model.

Two-Step Procedure

Step 1

Outer: Find $\hat{c}$ that optimizes Stein risk

Inner:

$$(\hat{\theta}(c), \hat{\lambda}(c))$$



ISP



Step 2

Debias:

$$\hat{\theta}_{\text{db}}$$



nominal coverage (θ^0)

Estimation

Bi-level Optimization and Shadow Price

- For fixed c , define the sample Lagrangian

$$\mathcal{L}_n(\theta, \lambda; c) = \phi_n(\theta) + \lambda(h(\theta) - c), \quad \lambda \geq 0.$$

- ▶ The pair $(\hat{\theta}(c), \hat{\lambda}(c))$ is characterized by the KKT system.
- ▶ The multiplier $\hat{\lambda}(c)$ has a shadow-price interpretation: it measures the marginal improvement in fit from relaxing the restriction.

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 - ▶ outer: estimate c that balances bias and variance,
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 - ▶ outer: estimate c that balances bias and variance,
 - ▶ inner: solve constrained problem for a fixed c .
- The outer step selects

$$\hat{c} \in \arg \min_{c \in \mathcal{C}} \hat{R}_{\text{lin}}(c), \quad \hat{\theta} = \hat{\theta}(\hat{c}), \quad \hat{\lambda} = \hat{\lambda}(\hat{c}).$$

Population Characterization and Active Region

- Let $(\theta^*(c), \lambda^*(c))$ denote the population KKT solution.
- If $\lambda^*(c) > 0$, the constraint binds; if $\lambda^*(c) = 0$, it is slack.

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- If $\lambda^*(c) > 0$, the constraint binds; if $\lambda^*(c) = 0$, it is slack.
- Define the active region

$$\mathcal{C}_{\text{active}} = \{c \in \mathcal{C} : \lambda^*(c) > 0\}.$$

- ▶ All profiling and ISP results are developed on $\mathcal{C}_{\text{active}}$, where the shadow-price system is nondegenerate.
- ▶ For $c \in \mathcal{C}_{\text{active}}$, $\theta^*(c) \neq \theta_0$; it is a pseudo-true parameter.

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- ▶ For $c \in \mathcal{C}_{\text{active}}$, $\theta^*(c) \neq \theta_0$; it is a pseudo-true parameter.
- On the active region, the KKT system is

$$\Phi_n(\theta, \lambda; c) = \begin{pmatrix} \nabla_{\theta} \phi_n(\theta) + \lambda \nabla_{\theta} h(\theta) \\ h(\theta) - c \end{pmatrix} = 0.$$

Theorem (Local Profiling Map)

Under smoothness and Jacobian nonsingularity, there exists a locally unique and smooth mapping

$$c \mapsto (\hat{\theta}(c), \hat{\lambda}(c)).$$

- This ensures the outer optimization over c is well posed.

Theorem (Joint Asymptotic Normality)

$$\sqrt{n} \begin{pmatrix} \hat{\theta}(c) - \theta^*(c) \\ \hat{\lambda}(c) - \lambda^*(c) \end{pmatrix} \xrightarrow{d} N \left(0, \begin{pmatrix} V_1(c) & V_2(c) \\ V_2(c)' & V_3(c) \end{pmatrix} \right).$$

- For fixed $c \in \mathcal{C}_{\text{active}}$, define

$$a(c) = \nabla_{\theta} h(\theta^*(c)), \quad A(c) = \nabla_{\theta\theta}^2 \mathcal{L}(\theta^*(c), \lambda^*(c); c).$$

- The key matrix is

$$M(c) = A(c)^{-1} - A(c)^{-1}a(c)(a(c)'A(c)^{-1}a(c))^{-1}a(c)'A(c)^{-1}.$$

- ▶ $a(c)$ is the *normal vector* to the binding constraint surface $\{\theta : h(\theta) = c\}$.
- ▶ It removes variation in the direction $a(c)$, preserving feasibility.
- ▶ Thus, $M(c)$ is a projected inverse Hessian: it restricts first-order variation to directions that satisfy the binding constraint.
- ▶ The variation of $\hat{\theta}(c)$ is projected to respect the binding constraint.

Outer Stein-type Risk Optimization

- Define the risk

$$R(c) = E[\|\hat{\theta}(c) - \theta_0\|^2].$$

- ▶ Using the decomposition

$$\hat{\theta}(c) - \theta_0 = (\hat{\theta}(c) - \theta^*(c)) + (\theta^*(c) - \theta_0)$$

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Local Bias Expansion

$$\theta^*(c) - \theta_0 = -\lambda^*(c)H_0^{-1}a_0 + o(\lambda^*(c)).$$

- This implies

$$R_{\text{lin}}(c) = \lambda^*(c)^2 a_0' H_0^{-1'} H_0^{-1} a_0 + \frac{1}{n} \text{tr}(V_1(c)).$$

- Thus, approximate via

$$\widehat{R}_{\text{lin}}(c) = \widehat{B}_{\text{lin}}(c) + \widehat{W}(c).$$

- ▶ $\widehat{B}_{\text{lin}}(c) = \hat{\lambda}(c)^2 \hat{a}' \hat{H}^{-1'} \hat{H}^{-1} \hat{a}.$
- ▶ $\widehat{W}(c) = \frac{1}{n} \text{tr}(\widehat{V}_1(c))$

Debiasing optimized estimator

- Let $c^\dagger = \arg \min_{c \in \mathcal{C}_{\text{active}}} R_{\text{lin}}(c)$.
- Even at $c^\dagger$, regularization bias remains:

$$\theta^*(c^\dagger) \neq \theta_0.$$

- Define

$$\hat{b}(c) = \hat{\lambda}(c) \hat{H}^{-1} \hat{a}, \quad \hat{\theta}_{\text{db}} = \hat{\theta}(\hat{c}) + \hat{b}(\hat{c}).$$

Theorem (Asymptotic Normality)

If $\hat{c} - c^\dagger = o_p(n^{-1/2})$, then

$$\sqrt{n}(\hat{\theta}_{\text{db}} - \theta_0) \xrightarrow{d} N(0, V_{\text{db}}(c^\dagger)).$$

- Debiasing restores correct centering for inference.

Inference

Individual Shadow Prices (ISP)

- The global shadow price is

$$\frac{\partial \mathcal{L}}{\partial c} = -\lambda.$$

- ▶ $-\lambda$ = marginal reduction in loss from relaxing the constraint
- ▶ i.e., λ = marginal gain in fit from relaxing the constraint

Individual Shadow Prices (ISP)

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Definition (Individual Shadow Price)

$$ISP_j(c) = 2\lambda(c) \operatorname{sign}(g_j(\theta(c))) [\Sigma^{-1}g(\theta(c))]_j.$$

- The use of a directional derivative ensures robustness to the sign of the restriction.
- Each ISP measures the marginal gain in fit from relaxing the j th restriction.
- ISP magnitudes therefore provide a natural measure of empirical relevance.
- This yields a unified, economically interpretable ranking of restrictions.

Asymptotic Distribution of the ISP Vector

Theorem (Asymptotic Normality of ISP)

Under regularity conditions and local stability of the sign map,

$$\sqrt{n}(\widehat{ISP}(c) - ISP^*(c)) \xrightarrow{d} N(0, \Sigma_{ISP}(c)).$$

- The variance reflects two sources:
 - ▶ uncertainty from $\hat{\theta}(c)$,
 - ▶ uncertainty from $\hat{\lambda}(c)$.
- Differences across ISPs reflect restriction-specific contributions within a common shadow-price system.
 - ▶ i.e., relative scale so that it is rankable.

Ranking Restrictions and Plateau Rule

- Rank restrictions using ordered ISP magnitudes.
- Under population separation, the estimated ranking is asymptotically correct.

Ranking Restrictions and Plateau Rule

- Rank restrictions using ordered ISP magnitudes.
- Under population separation, the estimated ranking is asymptotically correct.
- **Plateau Rule**
 - ▶ For each cutoff m , test for a structural break between upper and lower blocks.
 - ▶ Test a homogeneity condition within the lower block (plateau).
 - ▶ Select $\hat{m}(c)$ as the admissible break point with strongest evidence.
 - ▶ Restrictions below $\hat{m}(c)$: noise-like; Restrictions above $\hat{m}(c)$: signal

- Caveat

- ▶ In practice, the separability assumption for ISP magnitudes is not directly testable.
- ▶ When ISP values are small, their ordering can be sensitive to sampling variability.

Bootstrap Extension

- Caveat
 - ▶ In practice, the separability assumption for ISP magnitudes is not directly testable.
 - ▶ When ISP values are small, their ordering can be sensitive to sampling variability.
- A bootstrap procedure is therefore preferred because
 - ▶ captures variability,
 - ▶ provides a more reliable, data-driven determination of the cutoff.

Wild Bootstrap Consistency

Theorem (Wild bootstrap consistency)

Under regular conditions, for each fixed $c \in \mathcal{C}_{\text{active}}$, conditional on the data,

$$\sqrt{n} \begin{pmatrix} \hat{\theta}^{\mathcal{B}}(c) - \hat{\theta}(c) \\ \hat{\lambda}^{\mathcal{B}}(c) - \hat{\lambda}(c) \end{pmatrix} \xrightarrow{d^{\mathcal{B}}} \sqrt{n} \begin{pmatrix} \hat{\theta}(c) - \theta^*(c) \\ \hat{\lambda}(c) - \lambda^*(c) \end{pmatrix}.$$

Theorem (Wild bootstrap consistency for ISP)

Under regular conditions, for each fixed $c \in \mathcal{C}_{\text{active}}$, conditional on the data,

$$\sqrt{n}(\widehat{ISP}^{\mathcal{B}}(c) - \widehat{ISP}(c)) \xrightarrow{d^{\mathcal{B}}} N(0, \Sigma_{ISP}(c)).$$

- The bootstrap consistently reproduces the sampling distribution of the ISP vector, enabling inference on empirical relevance.

Bootstrap-based Plateau Inference

Bootstrap Algorithm

For each replication $b = 1, \dots, B$:

- generate pseudo-outcomes via multiplier-perturbed residuals,
- re-optimize to obtain $\hat{c}^{(b)}$,
- solve inner problem to obtain $(\hat{\theta}^{(b)}, \hat{\lambda}^{(b)})$,
- compute $ISP^{(b)}$.
- For each bootstrap draw, recompute the cutoff: $\{\hat{m}^{(b)}(c)\}_{b=1}^B$.

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-
- The bootstrap converts the untestable separation condition into observable variability.
 - ▶ Concentrated distribution: strong separation (clear signal vs. noise).
 - ▶ Diffuse distribution: weak separation (ambiguous ranking).
 - Summarize via mean and standard deviation to form an uncertainty-adjusted cutoff.

Simulation

Design

- We evaluate the finite-sample performance of the proposed framework using 1,000 Monte Carlo iterations with sample size $n = 1,000$.
- The parameter vector has dimension $1 + p = 11$, consisting of one intercept and $p = 10$ slope coefficients.
- The regressor matrix is

$$X = [1, X^{\text{raw}}], \quad X^{\text{raw}} = Z \Sigma_X^{1/2},$$

where Z has i.i.d. standard normal entries and Σ_X follows an AR(1) structure,

$$(\Sigma_X)_{j\ell} = \rho^{|j-\ell|}, \quad \rho = 0.8,$$

inducing strong dependence across covariates.

- The outcome is generated as

$$y = X\theta^0 + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2 I_n),$$

where σ_ε^2 is calibrated so that

$$\frac{\text{Var}(X\theta^0)}{\text{Var}(\varepsilon)} \approx 1,$$

corresponding to $R^2 \approx 0.5$.

Restrictions and Cases

- We consider the least-squares criterion

$$\phi_n(\theta) = \frac{1}{2n} \|y - X\theta\|^2.$$

- The credibility matrix is set to

$$\Sigma = I,$$

assigning equal weight to all restrictions and reflecting model uncertainty.

- The tolerance parameter satisfies

$$c \in (0, c_0], \quad c_0 = 1,$$

and is selected by minimizing the Stein-type risk criterion.

- We study three cases:

- ▶ **Case 1:** only structural restrictions are imposed.
- ▶ **Case 2:** an additional restriction,

$$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0,$$

is imposed and holds in the population.

- ▶ **Case 3:** the same additional restriction is imposed, but is misspecified and does not hold in the population.

Parameter Configurations

- The true parameter vector $\theta^0 \in \mathbb{R}^{11}$ consists of one intercept and ten slope coefficients.

- ▶ **Case 1:**

$$\theta_{\text{Case 1}}^0 = (0.1, 0.3, 0, -0.5, 0, 0.3, 0, 0.4, 0, -0.2, 0)'$$

- ▶ **Case 2:**

$$\theta_{\text{Case 2}}^0 = (0.1, 0.3, 0.2, -0.5, 0, 0.3, 0, 0.4, 0, -0.2, 0)',$$

so that

$$\theta_1^0 + \theta_2^0 + \theta_3^0 + \theta_4^0 = 0.$$

- ▶ **Case 3:**

$$\theta_{\text{Case 3}}^0 = (0.1, 0.3, 0, -0.5, 0, 0.3, 0, 0.4, 0, -0.2, 0)',$$

for which

$$\theta_1^0 + \theta_2^0 + \theta_3^0 + \theta_4^0 = -0.2 \neq 0.$$

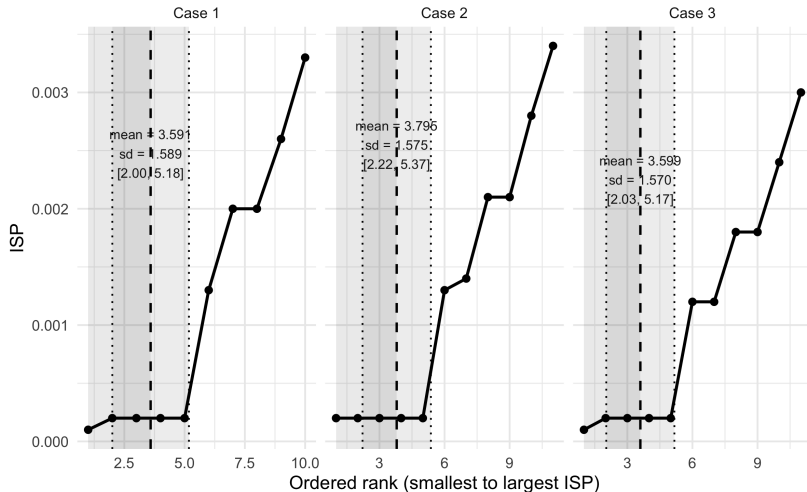
	Case 1			Case 2		Case 3	
Parameters ($j = 1, \dots, 10$)				$\theta_1 = 0, \theta_2 = 0, \dots, \theta_{10} = 0$			
Additional ($j = 11$)				$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$			
	N/A			<i>True</i>		<i>False</i>	
$\hat{\lambda}$	0.0034 (0.0008)			0.0035 (0.0009)		0.0031 (0.0008)	
True parameters	Estimates	ISP_j		Estimates	ISP_j	Estimates	ISP_j
θ_0^0	0.1031 (0.0161) [0.0031]			0.1035 (0.0182) [0.0035]		0.1031 (0.0161) [0.0031]	
θ_1^0	0.3028 (0.0269) [0.0028]	0.0020 (0.0005)		0.3031 (0.0304) [0.0031]	0.0021 (0.0006)	0.3028 (0.0269) [0.0028]	0.0018 (0.0005)
θ_2^0	-0.0051 (0.0341) [0.0051]	0.0002 (0.0001)	*	0.1942 (0.0386) [0.0058]	0.0013 (0.0003)	-0.0052 (0.0341) [0.0052]	0.0002 (0.0002)
θ_3^0	-0.4994 (0.0375) [0.0006]	0.0033 (0.0007)		-0.4993 (0.0425) [0.0007]	0.0034 (0.0008)	-0.4994 (0.0375) [0.0006]	0.0030 (0.0007)
θ_4^0	0.0026 (0.0377) [0.0026]	0.0002 (0.0001)	*	0.0029 (0.0427) [0.0029]	0.0002 (0.0002)	0.0025 (0.0377) [0.0025]	0.0002 (0.0001)
θ_5^0	0.3033 (0.0341) [0.0033]	0.0020 (0.0005)		0.3037 (0.0389) [0.0037]	0.0021 (0.0005)	0.3033 (0.0341) [0.0033]	0.0018 (0.0004)
θ_6^0	-0.0040 (0.0349) [0.004]	0.0002 (0.0001)	*	-0.0046 (0.0395) [0.0046]	0.0002 (0.0002)	-0.0040 (0.0350) [0.004]	0.0002 (0.0001)
θ_7^0	0.3988 (0.0327) [0.0012]	0.0026 (0.0006)		0.3986 (0.0372) [0.0014]	0.0028 (0.0007)	0.3987 (0.0327) [0.0013]	0.0024 (0.0006)
θ_8^0	-0.0019 (0.0319) [0.0019]	0.0002 (0.0001)	*	-0.0022 (0.0362) [0.0022]	0.0002 (0.0002)	-0.0019 (0.0319) [0.0019]	0.0002 (0.0001)
θ_9^0	-0.2003 (0.0322) [0.0003]	0.0013 (0.0004)		-0.2004 (0.0365) [0.0004]	0.0014 (0.0004)	-0.2003 (0.0322) [0.0003]	0.0012 (0.0004)
θ_{10}^0	0.0015 (0.0276) [0.0015]	0.0001 (0.0001)	*	0.0017 (0.0313) [0.0017]	0.0002 (0.0001)	0.0015 (0.0276) [0.0015]	0.0001 (0.0001)
$\theta_1^0 + \theta_2^0 + \theta_3^0 + \theta_4^0$	N/A			0.0002 (0.0001) *		0.0012 (0.0003)	
R^2	0.5060			0.5057		0.5060	
$\hat{c}$	0.6245 (0.0539)			0.6630 (0.0664)		0.6644 (0.0558)	
$\kappa(\hat{c})$	0.0001			0.0001		0.0001	
$\Delta(\hat{c})$	2.50E-05			9.00E-06		2.60E-05	

Nominal Coverage (Confidence level: 95%)

	Case 1		Case 2		Case 3	
	Analyt.	Boot.	Analyt.	Boot.	Analyt.	Boot.
Minimum	92.00	91.00	91.00	91.00	92.00	91.00
Q1	93.00	93.00	93.00	93.00	93.00	93.00
Median	94.00	94.00	94.00	94.00	94.00	94.00
Mean	93.82	93.91	93.91	93.91	93.91	93.91
Q3	94.50	95.00	95.00	95.00	94.50	95.00
Maximum	96.00	97.00	96.00	97.00	96.00	97.00

- Coverage remains close to the nominal 95% level in all scenarios.
- Analytical variance formulas perform comparably to the wild bootstrap.

- To assess the empirical relevance of each restriction, we rank the ISP values and apply the plateau rule.
- We use an uncertainty-adjusted cutoff based on the bootstrap distribution of the estimated cutoff.
 - ▶ Restrictions within the plateau are interpreted as empirically negligible;
 - ▶ those outside the plateau are interpreted as relevant for model fit.



- Restrictions classified in the plateau are those with small ISP magnitudes, indicating negligible empirical contribution to model fit.
- When the additional restriction is correct (Case 2), it is included in the plateau; when it is false (Case 3), it falls outside the plateau.

Main Findings

- The proposed estimator recovers θ^0 well across all cases, including the misspecified case.
 - ▶ The optimized tolerance and subsequent debiasing step effectively control the bias induced by possibly incorrect restrictions.
- Inference achieves close-to-nominal 95% coverage for θ^0 , with analytical and bootstrap-based procedures giving very similar results.
- The plateau rule successfully separates empirically irrelevant restrictions from relevant ones.
 - ▶ Restrictions classified in the plateau are those with small ISP magnitudes, indicating negligible empirical contribution to model fit.
 - ▶ When the additional restriction is correct (Case 2), it is included in the plateau; when it is false (Case 3), it falls outside the plateau.

Empirical Application

Motivation

- We apply the framework to the Solow growth model, where theory implies structural restrictions on the relationships among growth determinants.
- Motivation
 - ▶ Growth regressions face a tradeoff between theoretical discipline and empirical flexibility
 - ★ Growth processes may exhibit nonlinearities arising from heterogeneous dynamics or multiple equilibria (Durlauf and Johnson, 1995; Temple, 1999)
 - ▶ Exact restrictions may be too strong in heterogeneous macroeconomic data
 - ★ Distinct structural mechanisms—such as convergence clubs, stratification, or polarization—can generate observationally similar reduced-form relationships (Quah, 1996; Durlauf and Quah, 1999)

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- Our approach
 - ▶ Incorporates Solow-model structure within estimation
 - ▶ Allows flexible functional forms (e.g., polynomial terms for illustrative purpose)
 - ▶ Accommodates restriction misspecification via soft constraints
 - ▶ Quantifies the empirical relevance of theoretical restrictions

Results

	Textbook Solow		Our estimation
	Unrestricted	Restricted	
constant	4.651 (0.0010)	8.3748 (0.0000)	4.1493 (2.4794)
$\ln s$	1.2756 (0.0000)	1.3791 (0.0000)	1.2674 (0.1046)
$\ln(n + 0.05)$	-2.7087 (0.0000)	-1.3791 (0.0000)	-2.8923 (0.9532)
Restrictions	Wald		Plateau
$\theta_s + \theta_n = 0$	4.427 (0.0380)		*
$\theta_s - (-\theta_n)^2 = 0$			*
$\theta_s - (-\theta_n)^3 = 0$			
R^2	0.6272	0.6084	0.6269
$\hat{c}$ ($c_0 = 200$)		200	
Risk estimate		1.7797	
Bias proxy		1.1703	
Variance proxy		0.0694	

Conclusion

• Motivation and questions

- ▶ Model uncertainty arising from multiple, potentially misspecified restrictions
- ▶ Questions:
 - ★ How to estimate consistently under model uncertainty?
 - ★ How to assess the empirical relevance of restrictions?

• Our framework

- ▶ Estimation: data-driven optimization of the tolerance parameter and debiasing of the constrained estimator
- ▶ Inference: individual shadow prices with bootstrap-based plateau rule for separating signal from noise
- ▶ Simulation: consistent estimation, recovery of nominal coverage, and improved performance of the bootstrap procedure

• Implications

- ▶ Robust inference under model uncertainty and restriction misspecification
- ▶ Data-driven incorporation of restrictions from economic theory